
Project / Lab Reference Deck

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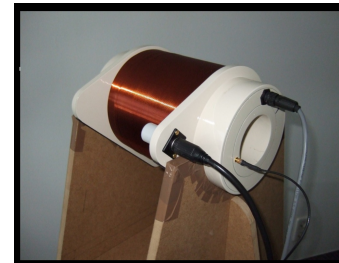
Personal: jackynam@outlook.com



MEDICAL IMAGING LAB WORK

Experimental and diagnostic work performed with the following systems:

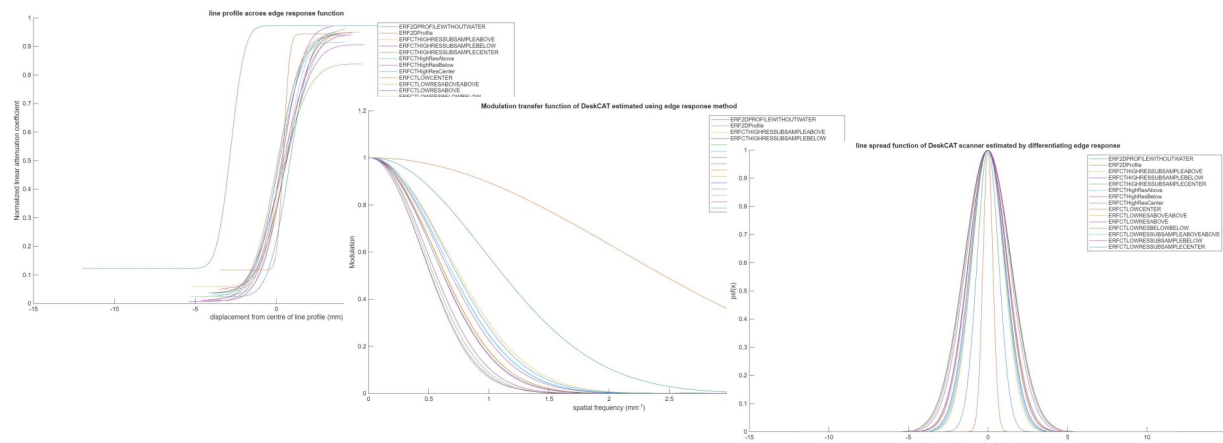
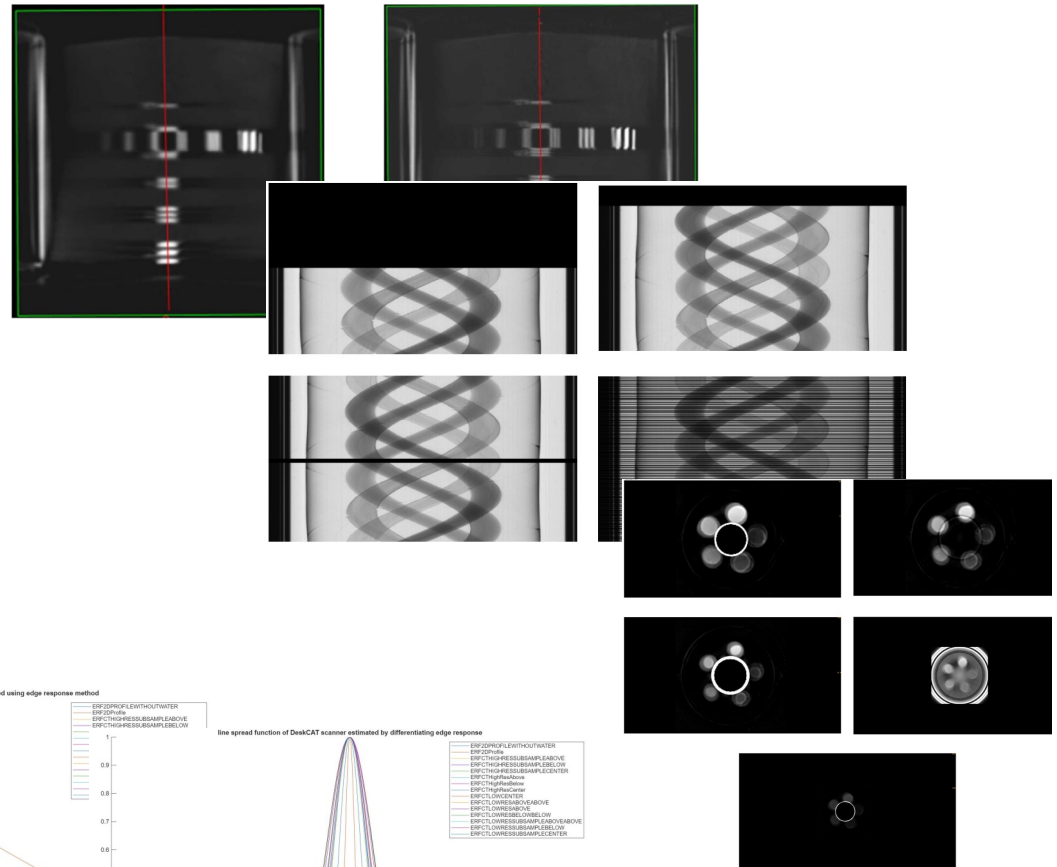
- [DeskCAT](#) for simulated X-RAY and CT
- Clarius [L15](#) and [C3](#) for Ultrasound
- Earth's Field MRI apparatus for NMR
- [0.345T ilumr](#) system for MRI



Applied Fourier transform theory to medical image formation, reconstruction, and analysis across multiple imaging modalities using MATLAB/Python scripting. (CT/X-Ray shown on a phantom)

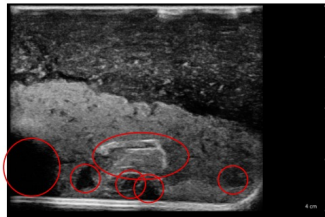
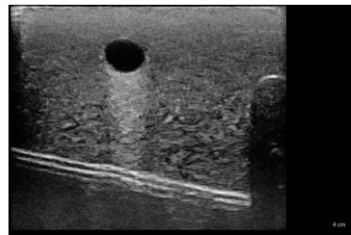
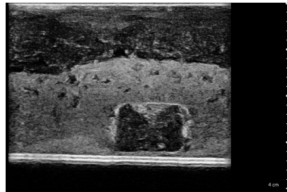
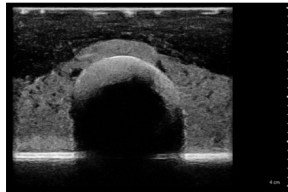
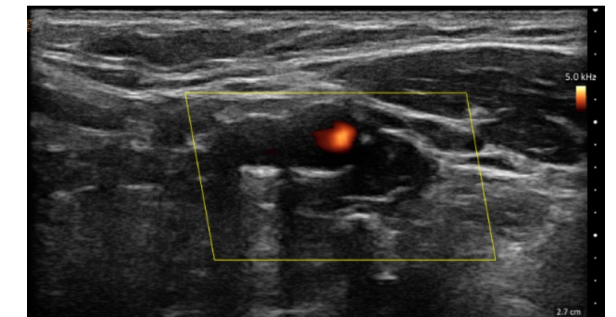
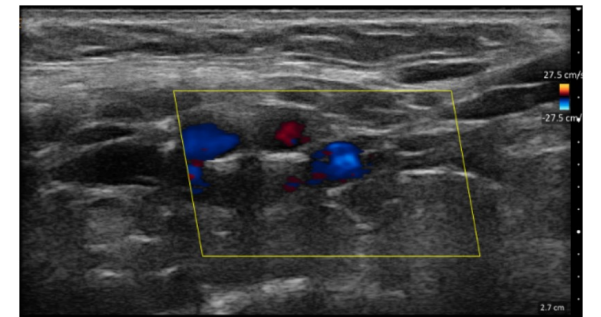
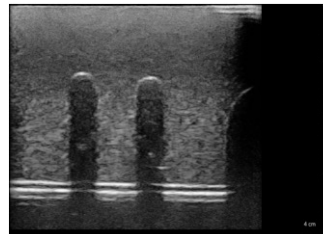
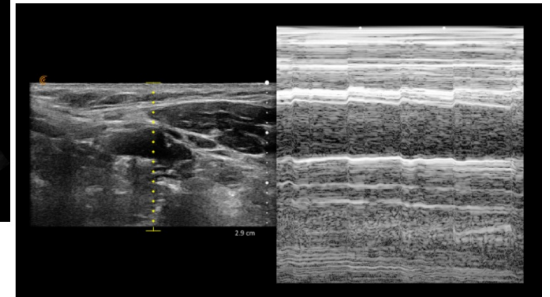
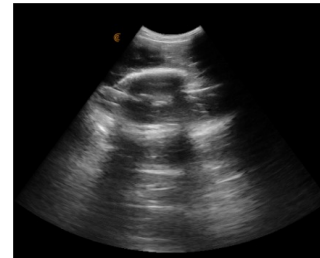
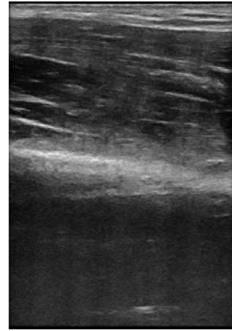
Analyzed sources of noise, artifacts, and resolution limits in X-ray, CT.

Evaluated image quality metrics (PSF, MTF) and QA/QC phantom calibration standards for clinical imaging systems.



Conducted hands-on ultrasound imaging and elastography using handheld Bluetooth-connected US systems on iPads.

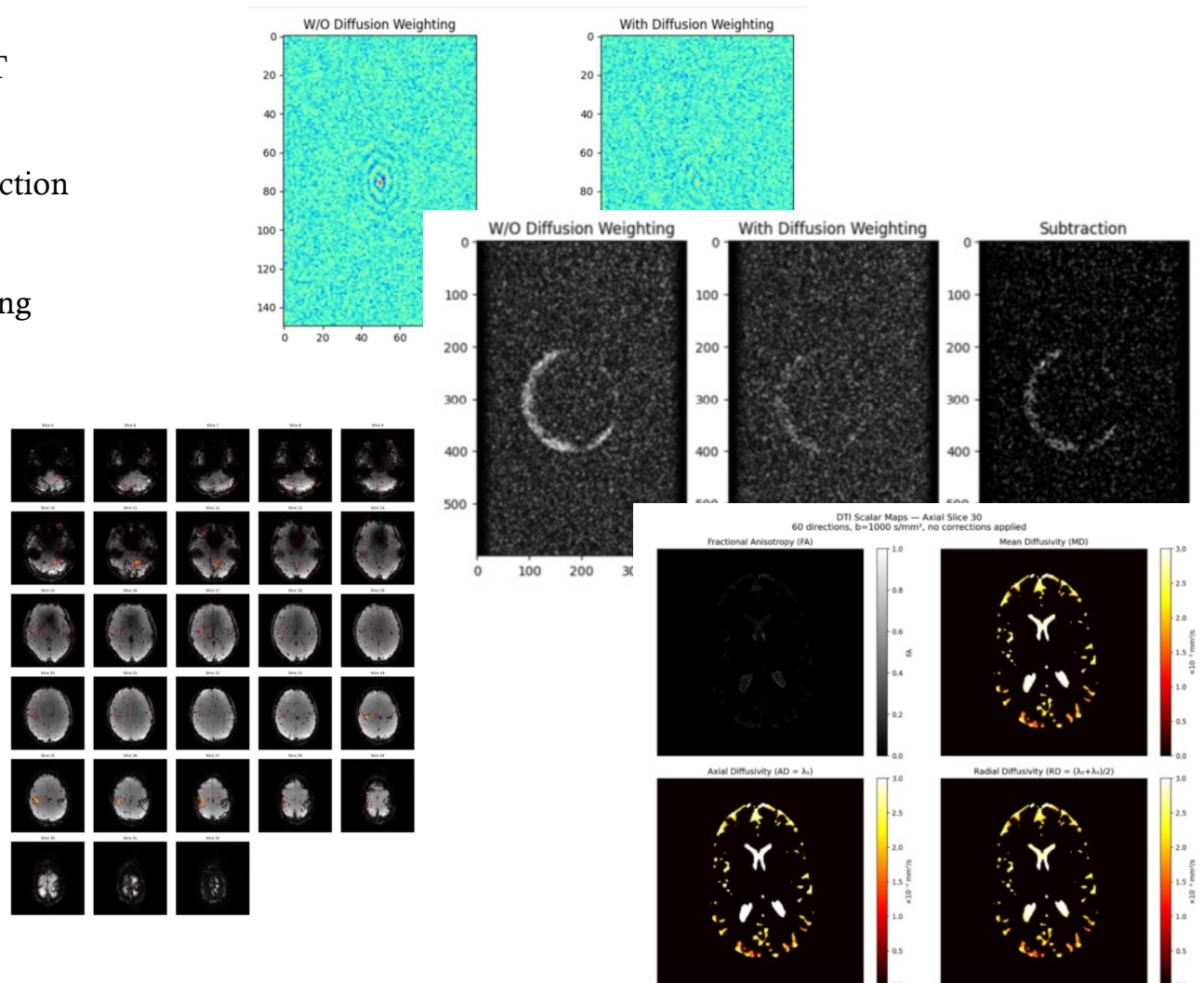
Analyzed sources of noise, artifacts, and resolution limits in ultrasound.



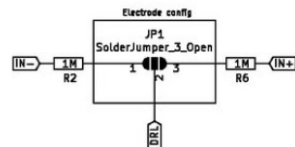
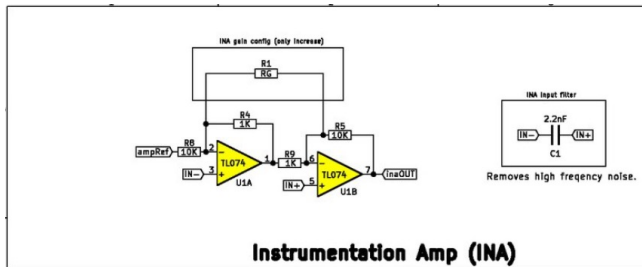
Performed MRI experiments on 0.33T
Flowerpot MRI systems including
calibration, T1/T2 measurement,
shimming, and imaging sequence selection
(gradient echo, spin echo, RARE).

Analyzed advanced MRI data including
diffusion weighted imaging (DWI),
diffusion tensor imaging (DTI), and
functional MRI (fMRI) from human
scanning sessions.

Reconstructed in Python.

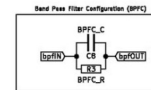
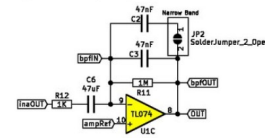
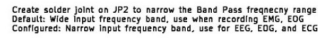


2025 – REM SLEEP EEG



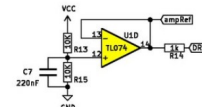
Bridge all three pads of the solder jumper to configure for 2 Electrode operations instead of the 3 electrodes
Note: 2 electrodes will give noisier output than 3 electrodes!

Electrode reference

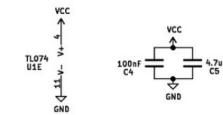


C8 (BPFC_C) & R3 (BPFC_R) on the back side of the PCB can be used to configure the bandpass filter. Use them to configure output Gain (decrease) & Band (frequency range).

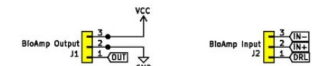
1000x Gain (Bandpass)



Amp Ref + Driven Right Leg (DRL)



Power supply filtering

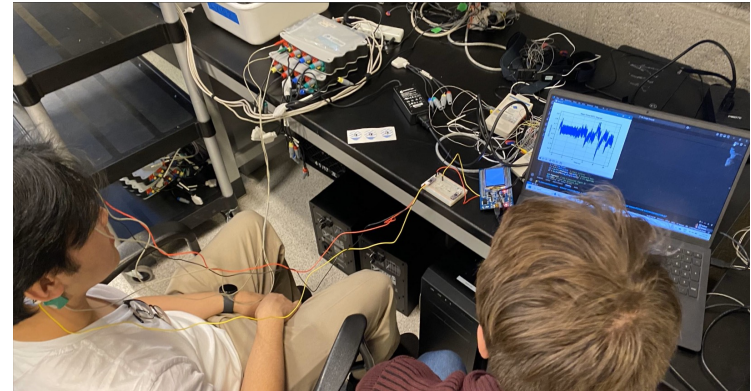


Header pins / Connectors

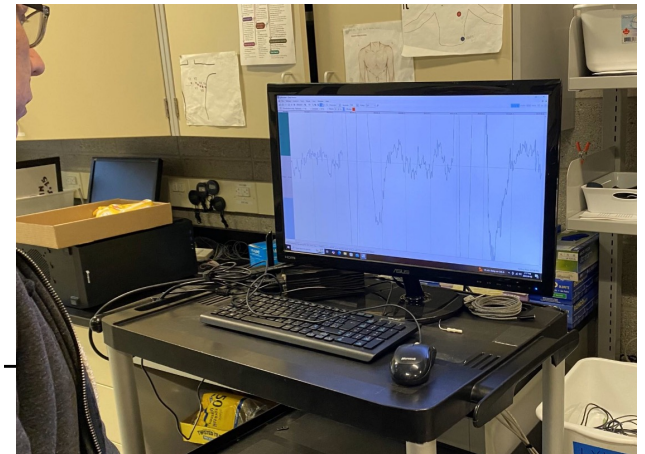
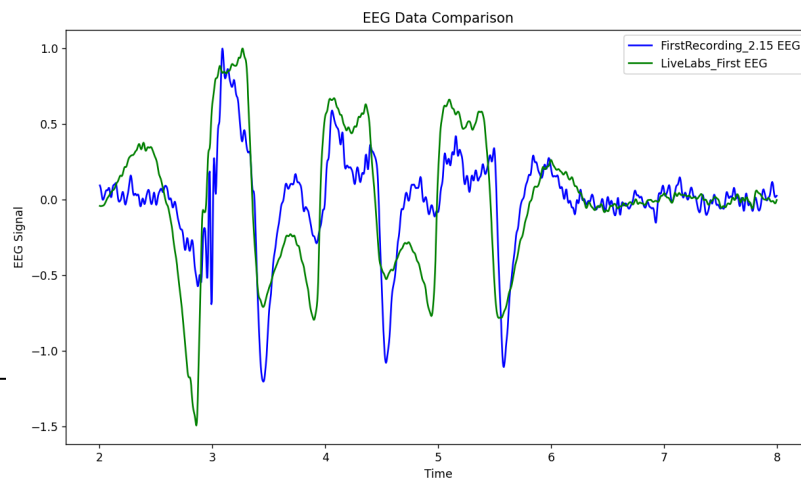
Hardware used sourced from [BioAmp EXG](#)
[Pill](#) used with documentation from
[OpenEEG](#).

2025 – REM SLEEP EEG

Clinical reference EEG
equipment supplied and tested
at [McMaster LiveLAB](#)



```
===== First Recording =====  
EEG Data Minute 0  
Raw Spectrum R²: 0.4818934173922921  
Envelope Spectrum R²: 0.9661411392288032  
Cum Sum Spectrum R²: 0.9932037351313463  
EEG Data Minute 1  
Raw Spectrum R²: 0.5088660188246688  
Envelope Spectrum R²: 0.7430213020803893  
Cum Sum Spectrum R²: 0.8773447508242305  
EEG Data Minute 2  
Raw Spectrum R²: 0.4736143094498777  
Envelope Spectrum R²: 0.9392312325333119  
Cum Sum Spectrum R²: 0.9967189426464305  
EEG Data Minute 3  
Raw Spectrum R²: 0.5479455184358688  
Envelope Spectrum R²: 0.7628989520024941  
Cum Sum Spectrum R²: 0.775161612600471  
  
===== Second Recording =====  
EEG Data Minute 0  
Raw Spectrum R²: 0.5343477230401192  
Envelope Spectrum R²: 0.9847871910660738  
Cum Sum Spectrum R²: 0.9994075701595474  
EEG Data Minute 1  
Raw Spectrum R²: 0.48244575691037406  
Envelope Spectrum R²: 0.945509804021429  
Cum Sum Spectrum R²: 0.9830070405271604  
EEG Data Minute 2  
Raw Spectrum R²: 0.535920299506134  
Envelope Spectrum R²: 0.9671612619510352  
Cum Sum Spectrum R²: 0.9976695150274284  
EEG Data Minute 3  
Raw Spectrum R²: 0.6721621683522725  
Envelope Spectrum R²: 0.9136236705912524  
Cum Sum Spectrum R²: 0.9071806724738036
```



2022 – HIP IMPLANT

Addressed concerns of a 51-year-old male after a left total hip arthroplasty.

Symptoms presented included:

- Increasing pressure on hip affecting mobility
- Excessive increase in weight (lack of exercise, poor diet, quitting smoking)
- Arthritis in hip causing little to no pain with occasional flare-ups

5 Years After Arthroplasty:

- Persistent, slight pain in thigh and groin

7 Years After Arthroplasty:

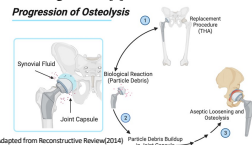
- Severe pain radiating towards the hip
- Shortness of breath after light exercise
- Slight gait disturbance

Diagnosed with aseptic periprosthetic osteolysis, designed a biocompatible implant – able to sustain lifestyle and exercise patterns.



Background: Arthritis and Osteolysis

- Synovial fluid:** natural lubricating fluid allowing for smooth joint movement protecting bones and cartilage [1]
- Inability of the synovial joint fluid to naturally regenerate leads to cartilage degradation and inflammation of the joints, causing arthritis [1]



- A lack of seal on the bone-implant interface and friction can lead to particulate debris from the implant causing inflammatory biological reactions [2]
- Wear on bearing surfaces releases particles into the joint capsule which migrate to the tissues [2]
- Foreign response to particle debris initiates bone loss

Ke Huy-Quan: Profile & Diagnosis

Fast Facts: 51 years old, 111kg, 181cm, BMI = 33.9

History & Symptoms

Prior to Arthroplasty:

- Excessive weight gain (lack of exercise, poor diet) putting pressure on hip & restricting mobility
- End-stage arthritis in the hip causing minor pain

Cementless implant with polyethylene surface is inserted without complications

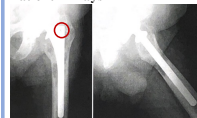
5-7 Years after Arthroplasty:

- Pain in hip & groin becoming severe, radiating
- Shortness of breath after exercise & gait disturbance

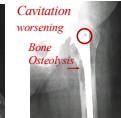
Diagnosis: Aseptic Periprosthetic Osteolysis of previous hip replacement

- Inflammatory response to particles excreted by polyethylene implant causing implant degradation and aseptic loosening
- Explains the implant's cavitation, as well as implant's resulting weakness

Patient X-Rays



Left hip 5 years post-replacement



7 years post-replacement

Summary

- The Patient - Ke Huy-Quan - underwent a total hip arthroplasty due to end-stage arthritis, later developing **aseptic periprosthetic osteolysis** of the implant due to his body's reaction to particulate debris caused by wear
- Thus, his new implant must be made of **unreactive** biocompatible materials, and would support high levels of **force** resulting from his increased weight gain
- Our implant uses novel **layered materials** to strike balances between **biocompatibility & wear resistance** (in the femoral head and acetabular cup), and **ultimate strength and bone density reduction** (in the stem)
- A **spring** connects the femoral head to the stem, removing **mechanical load and absorbing shock**

Materials

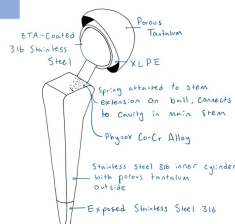
COMPONENT	HIGHLY CROSS-LINKED POLYETHYLENE + XLPE ZIRCONIA TOUGHENED ALUMINA + ZTA				
	STAINLESS STEEL	POROUS TANTALUM	XLPE	ZTA	PHYNOX CO-OR ALLOY
Acetabular Cup		✓	✓		
Femoral Head	✓			✓	
Femoral Stem	✓	✓			
Spring					✓

Coatings and Combinations:

- 316 Stainless Steel interior of femoral stem coated with Porous Tantalum of high ultimate strength
- Zirconia Toughened Alumina acts as a protective coating on the femoral head
- Acetabular cup is lined with Highly Crosslinked Polyethylene against Porous Tantalum cup.

Spring:

- Cobalt Chromium alloy** with high corrosion resistance and biocompatibility [3]



Metal on Polyethylene:

- Metal 316 Stainless Steel** Femoral stem paired with an **Ultra High Cross-Linked Polyethylene Liner**

Justification

Metals:

Porous Tantalum

- Preserves bone strength and density over long term recovery periods
- Nearly identical elastic modulus to accurately mimic bone stiffness – less bone loss [4]

Stainless Steel 316L

- Compensates for the lack of initial stiffness of Porous Tantalum with a high elastic modulus > Ti
- Long lasting reliability in biocompatibility and corrosion resistance [5]

Coatings:

XLPE – Highly Crosslinked Polyethylene

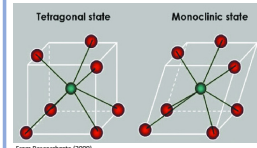
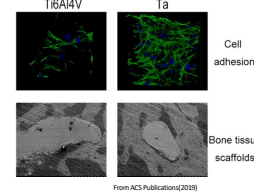
- 1200GPa ultimate strength with reduced particle debris
- Crosslinked property minimizes wear, reducing inflammation

ZTA – Zirconia Toughened Alumina

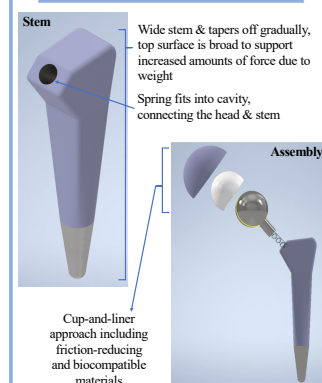
- Resistant to an increase in corrosion and has a low brittleness, reducing fracture
- High elastic modulus, limiting implant stress [6]

Spring and Phynox:

- Absorbs shock from compressive stress put on implant by removing mechanical load [7]
- A stiffer spring will account for increased weight, decreasing compression
- Extremely high corrosion wear and fatigue life [7]



CAD Model



Surgical Considerations

Prior failed THA to be managed and resolved

- Bone degraded from osteolysis removed - process decreases overall bone mass

Implant Fixation

- Cemented Implant mimics natural bone – keeps implant stable in the area of reduced bone mass that is not sufficient to hold the implant without support [8]

Surgical Changes for an obese patient [9]

Pre-op Recommendations	Weight loss
Operative Changes	Posterior approach, bariatric equipment, IV antibiotics
Post-op Recommendations	Higher nursing vigilance, epidural anesthetics, physiotherapy

Sources

- "Arthritis and Joint Pain Center: Symptoms, causes, types, and treatment." BAAAD. [Online]. Available: <https://www.baaad.ca/en/condition/arthritis-and-joint-pain/>
- "What is Arthritis?" American Medical Association. [Online]. Available: <https://www.ama-assn.org/practicing/conditions-and-diseases/arthritis>
- "Arthritis." Arthritis Research UK. [Online]. Available: <https://www.arthritisresearchuk.org/>
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